Seminar 2 · Instructor solutions

Three problems · Consumer choice, KKT and Roy

Suggested 90-minute route. Roy recap 10 minutes; problem 1 about 20; problem 2 about 25; problem 3 about 30; final questions 5. The extra Roy calculation in 2(c) and full boundary correspondences in 3(d) can be reserve. The student sheet has three problems, not a slide presentation. It prepares the methods used in Homework 1 without reproducing its parameterisations.

What else to reinforce. Existence is separate from uniqueness; KKT necessity is separate from sufficiency. Check all four groups and all regime boundaries. Multipliers depend on the utility scale; Roy ratios do not under a differentiable increasing transformation with positive derivative. A minimum of two affine utilities is handled by an auxiliary-variable problem, not by inventing a derivative at its kink.

Opening: Roy

At a regular consumer optimum the envelope formulas give $v_m = \lambda$ and $v_{p_i} = -\lambda x_i^*$. Taking the negative ratio removes the utility units. The direct fixed-bundle proof from the lecture also applies when the direct index is nonsmooth, as in problem 3. Ask why a value computed only at fixed numerical prices cannot provide the missing general price derivatives. After normalising the numeraire price to 1, Roy with respect to r recovers q, and budget exhaustion recovers a.

1. A head start changes the corner

(a) Representation and existence. Use $g = \log u = \log(1 + x_1) + \log(1 + x_2)$. The original index is positive, so log is strictly increasing on its range. The Hessian of g is diagonal with entries $-1/(1 + x_i)^2 < 0$: strict concavity gives uniqueness on a convex budget. Continuity and positive prices give existence on a nonempty compact budget, and positive partial derivatives imply budget exhaustion. The original product index is not concave: its Hessian has off-diagonal entries 1 and zero diagonal, hence a positive and a negative eigenvalue. Do not use the product's curvature as the sufficiency argument.

(b) Fixed prices and all incomes. For the log representation,

$$\mathcal{L} = g + \lambda(m - x_1 - 2x_2) + \mu_1 x_1 + \mu_2 x_2.$$

Stationarity:

$$\frac{1}{1 + x_1} - \lambda + \mu_1 = 0, \qquad \frac{1}{1 + x_2} - 2\lambda + \mu_2 = 0.$$

Include $x_i \geq 0$, $m - x_1 - 2x_2 \geq 0$, $\lambda, \mu_i \geq 0$, and all three complementary-slackness products.

At an interior optimum, $1 + x_1 = 2(1 + x_2)$ and $x_1 + 2x_2 = m$. Thus

$$x_1 = \frac{m + 1}{2}, \quad x_2 = \frac{m - 1}{4}, \quad \lambda = \frac{2}{m + 3},$$

valid for $m > 1$. At a corner $x_2 = 0$, we have $x_1 = m$, $\lambda = 1/(1+m)$, $\mu_1 = 0$ and $\mu_2 = (1-m)/(1+m)$. These are valid for $0 < m \leq 1$.

m	x_1	x_2	λ	μ_1	μ_2
$1/2$	$1/2$	0	$2/3$	0	$1/3$
1	1	0	$1/2$	0	0
5	3	1	$1/4$	0	0

The original g is strictly concave, so these KKT candidates are sufficient and unique. At m=1 the zero quantity has zero bound multiplier.

(c) Roy. Write $S = m + p_1 + p_2$. Then

$$(v_{\mathsf{log}})_m = 2/S, \qquad (v_{\mathsf{log}})_{p_i} = 2/S - 1/p_i.$$

Roy yields

$$x_i^* = \frac{S}{2p_i} - 1,$$

positive for both goods iff $m > |p_1 - p_2|$. At prices (1,2) this reproduces the interior solution above. One derivation of the supplied value uses $z_i = 1 + x_i$: when both lower bounds $z_i \geq 1$ are slack, split effective wealth S equally across the two log terms.

For the raw index $v_{\mathsf{raw}} = e^{v_{\mathsf{log}}}$, both relevant derivatives acquire the same positive factor $e^{v_{\mathsf{log}}}$, so the Roy ratio is unchanged. The raw budget multiplier is instead $u(x^*)$ times the log multiplier; it is not numerically invariant.

2. Spend it or keep it

(a) Reduction and regimes. Continuity on the compact positive-price budget gives existence. Utility is increasing in a, so $a = m - rq$. The reduced problem is

$$\max_{0 \le q \le m/r} \{m - rq + 4\sqrt{1+q}\}.$$

Its derivative is $-r + 2/\sqrt{1+q}$ and its second derivative is $-(1+q)^{-3/2} < 0$. The stationary quantity is $4/r^2 - 1$, giving

$$q^* = \min\left\{\frac{m}{r}, \max\left\{\frac{4}{r^2} - 1, 0\right\}\right\}, \qquad a^* = m - rq^*.$$

The unique optimum is

$$(a^*, q^*) = \begin{cases} (m, 0), & r \ge 2, \\ (0, m/r), & 0 < r < 2, \quad m \le 4/r - r, \\ (m - 4/r + r,\ 4/r^2 - 1), & 0 < r < 2, \quad m > 4/r - r. \end{cases}$$

At r=2, the zero quantity is still uniquely optimal: the reduced derivative is zero only at q=0 and negative afterwards. At m=4/r-r the two positive-q formulas agree, with a=0; there is no multiplicity.

(b) KKT verification. Use $\mathcal{L} = u + \lambda(m - a - rq) + \mu_a a + \mu_q q$. Stationarity is

$$1 - \lambda + \mu_a = 0, \qquad \frac{2}{\sqrt{1+q}} - \lambda r + \mu_q = 0.$$

Together with primal/dual feasibility, require $\lambda(m - a - rq) = \mu_a a = \mu_q q = 0$. The direct index is concave (affine in a, strictly concave in q); constraints are affine, so these conditions are sufficient. Uniqueness follows from the strictly concave reduced objective and exhaustion, not strict concavity in the full pair.

r	m	a	q	λ	μ_a	μ_q
1	1	0	1	$\sqrt{2}$	$\sqrt{2} - 1$	0
1	6	3	3	1	0	0
3	6	6	0	1	0	1

In general, lambda is 1 in the no-service and both-goods regimes. In the all-service regime it is $2/[r\sqrt{1 + m/r}] \ge 1$, with $\mu_a = \lambda - 1$, $\mu_q = 0$. In the no-service regime $\mu_q = r - 2$.

(c) Value and Roy.

$$v(r, m) = \begin{cases} m + 4, & r \ge 2, \\ 4\sqrt{1 + m/r}, & 0 < r < 2, \quad m \le 4/r - r, \\ m + r + 4/r, & 0 < r < 2, \quad m > 4/r - r. \end{cases}$$

In the both-goods region $v_m = 1$ and $v_r = 1 - 4/r^2$, so $-v_r/v_m = 4/r^2 - 1 = q^*$. In the all-service region,

$$v_m = \frac{2}{r\sqrt{1 + m/r}}, \qquad v_r = -\frac{2m}{r^2\sqrt{1 + m/r}}, \qquad -\frac{v_r}{v_m} = \frac{m}{r}.$$

In the no-service region the ratio is zero. Recover a by exhaustion with unit numeraire price. Values and first derivatives agree at the regime boundaries, so Roy remains available there even though the quantity rules have kinks.

(d) Income effects and a zero multiplier. At r=1, q=min(m,3), a=max(m-3,0). Below 3 additional income buys q; above 3 it buys only a. At m=3 the bundle is (0,3), lambda=1 and $\mu_a = 0$. A binding lower bound does not imply a strictly positive multiplier.

3. Keep both critics happy

(a) Reformulation. Each feasible augmented triple has $t \le \min\{2x_1 + x_2, x_1 + 2x_2\} = u(x)$. For each original bundle one can set t=u(x); hence maximising t is equivalent. The objective is linear and every constraint is affine, so concave-program KKT sufficiency applies without differentiating the min index. The variable t is unrestricted; a constraint t>=0 is unnecessary. At positive wealth the optimum has t>0 anyway. Adding this redundant bound would preserve the optimum, but its multiplier would be zero there.

The Lagrangian is

$$\begin{aligned}\mathcal{L} = t &+ \lambda(m - p_1x_1 - p_2x_2) \\ &+ \eta_1(2x_1 + x_2 - t) + \eta_2(x_1 + 2x_2 - t) + \mu_1x_1 + \mu_2x_2.\end{aligned}$$

Stationarity:

$$1 - \eta_1 - \eta_2 = 0, \quad 2\eta_1 + \eta_2 - \lambda p_1 + \mu_1 = 0, \quad \eta_1 + 2\eta_2 - \lambda p_2 + \mu_2 = 0.$$

Include nonnegative lambda, eta and mu, every primal inequality, and complementary slackness:

$$\lambda(m - p \cdot x) = 0, \quad \eta_1(2x_1 + x_2 - t) = 0, \quad \eta_2(x_1 + 2x_2 - t) = 0, \quad \mu_i x_i = 0.$$

(b) Equal prices. At p=(1,1), $x_1 = x_2 = m/2$, $t = 3m/2$. Multipliers are $\lambda = 3/2$, $\eta_1 = \eta_2 = 1/2$ and $\mu_1 = \mu_2 = 0$. They satisfy KKT, hence give a global optimum. For uniqueness, $u(x) \le [3(x_1 + x_2)]/2 \le 3m/2$; equality in the first inequality requires the two scores equal, i.e. x1=x2, and the budget fixes both at m/2.

(c) General prices. Write $S = p_1 + p_2$. At a balanced candidate,

$$x_1 = x_2 = m/S, \quad t = 3m/S, \quad \mu_1 = \mu_2 = 0.$$

Stationarity gives

$$\lambda = 3/S, \qquad \eta_1 = \frac{2p_1 - p_2}{S}, \qquad \eta_2 = \frac{2p_2 - p_1}{S}.$$

Both eta multipliers are nonnegative iff $1/2 \le p_1/p_2 \le 2$. When the inequalities are strict, both score constraints bind, giving a unique balanced optimum.

If $p_1/p_2 < 1/2$, choose $(x_1, x_2) = (m/p_1, 0)$ and t=m/p1. Set $\eta_1 = 0$, $\eta_2 = 1$, $\lambda = 1/p_1$, $\mu_1 = 0$ and $\mu_2 = p_2/p_1 - 2 > 0$. If $p_1/p_2 > 2$, choose $(0, m/p_2)$, t=m/p2, with $\eta_1 = 1$, $\eta_2 = 0$, $\lambda = 1/p_2$, $\mu_1 = p_1/p_2 - 2 > 0$, $\mu_2 = 0$. These satisfy the full KKT system. Strict signs and positive prices establish the corresponding unique corners.

An alternative argument reduces the exhausted budget to one dimension; the minimum of the two affine score functions is piecewise linear, and its optimum is at an endpoint or at their intersection, with a flat segment only at the two boundary ratios.

(d) Roy and boundary sets. In the strict balanced region, $v = 3m/S$, $v_m = 3/S$ and $v_{p_i} = -3m/S^2$. Roy recovers x1=x2=m/S. More generally,

$$v(p, m) = m \max\left\{\frac{1}{p_1}, \frac{1}{p_2}, \frac{3}{p_1 + p_2}\right\}.$$

At $p_1 = p_2/2$, the first and third branches tie and have different price gradients. At $p_1 = 2p_2$, the second and third do. Hence v is not differentiable in prices there; Roy cannot choose a unique point from the full optimal set.

For the extension, the demand correspondences at the boundaries are

$$\begin{aligned} p_1 = p_2/2: &\quad D = \{x \geq 0 : p \cdot x = m,\ x_1 \geq x_2\}, \\ p_1 = 2p_2: &\quad D = \{x \geq 0 : p \cdot x = m,\ x_1 \leq x_2\}. \end{aligned}$$

Each is the line segment between the appropriate corner and the balanced bundle. Every point is optimal. At the first boundary the multipliers $\eta_1 = 0$, $\eta_2 = 1$, $\lambda = 1/p_1$, mu=0 support the whole segment; at the second swap the roles. This is a useful final illustration: convex preferences can have unique demands for many prices without being strictly convex, and boundary prices can expose an entire segment of optimisers.